



Heaps, Homework 7

1 Heaps.

1. Is it possible to extract the min/max element of a heap while keeping the heap a heap, in constant time (in the comparison model)? Prove or disprove it.
2. Is the heap sort a stable sort? Prove or give a counter example.
3. Prove that heap construction (Build-Heap) takes $O(n)$ time.

```
input: Array  $H = H_1..H_n$   
1  $i \leftarrow \lfloor \frac{1}{2}n \rfloor$   
2 while  $i \geq 1$  do  
3   | Heapify-down ( $H, i$ )  
4   |  $i \leftarrow i - 1$   
5 end  
6 return  $H_1..H_n$ 
```

Algorithm 1: Build a heap.

4. Given an array where all its elements are different from each other, write an algorithm that returns the k largest elements. Prove its correctness and bound its running time.